

Spiral Similarity

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Thank Aatman for getting neat diagrams everyone.

Definition (Complete Quadrilateral). Given a quadrilateral $ABCD$ (possibly convex), let $E = AB \cap CD$, $F = AD \cap BC$. $ABCDEF$ is now called a complete quadrilateral. Thus, a complete quadrilateral $ABCDEF$ is determined by 4 lines and their 6 unique intersections.

§1 Spiral Similarity

A spiral similarity is the combination of a rotation and a dilation from some center.

Lemma 1.1 (Spiral Centers)

Let $\overline{AB}$ and $\overline{CD}$ be segments. Then, let $E = AC \cap BD$. Then, $O = (ABX) \cap (XCD)$ is the unique center of spiral similarity that takes A to C and B to D .

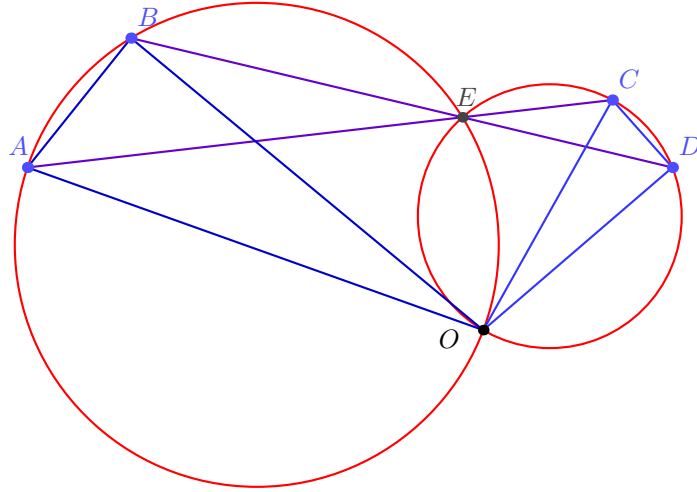


Figure 1: O is the unique center of spiral similarity that takes A to C and B to D

Proof. First, we show that O works by angle chase $\angle OAB = \angle OEB = \angle OED = \angle OCD$. Similarly, $\angle OBA = \angle OEA = \angle OEC = \angle ODC$. Thus, $\triangle OAB \sim \triangle OCD$.

The uniqueness can be shown by assuming that another center of spiral similarity exists and following this argument in reverse. $\square$

Lemma 1.2

For every quadruple of points A, B, C, D , there exists a unique spiral similarity that creates the two maps- $A \mapsto C, B \mapsto D$ unless $ABDC$ is a parallelogram.

Proof. Follows from Lemma 1.1 □

Lemma 1.3 (Spiral Similarities come in pairs)

The the center of spiral similarity of $AB \mapsto CD$ is also the spiral center that takes $AC \mapsto BD$.

Proof. Try showing the result using angle chase. □

Theorem 1.4 (Miquel Point)

Given a complete quadrilateral $ABCDEF$, the circles $(EAB), (EBC), (FAB), (FCD)$ concur at a point M known as the Miquel Point M . M is now the center of six spiral similarities or three pairs of spiral similarities.

- $\overline{AB} \leftrightarrow \overline{CD}$ and $\overline{AD} \leftrightarrow \overline{BC}$
- $\overline{FD} \leftrightarrow \overline{DC}$ and $\overline{FB} \leftrightarrow \overline{DE}$
- $\overline{FA} \leftrightarrow \overline{CE}$ and $\overline{FC} \leftrightarrow \overline{AE}$

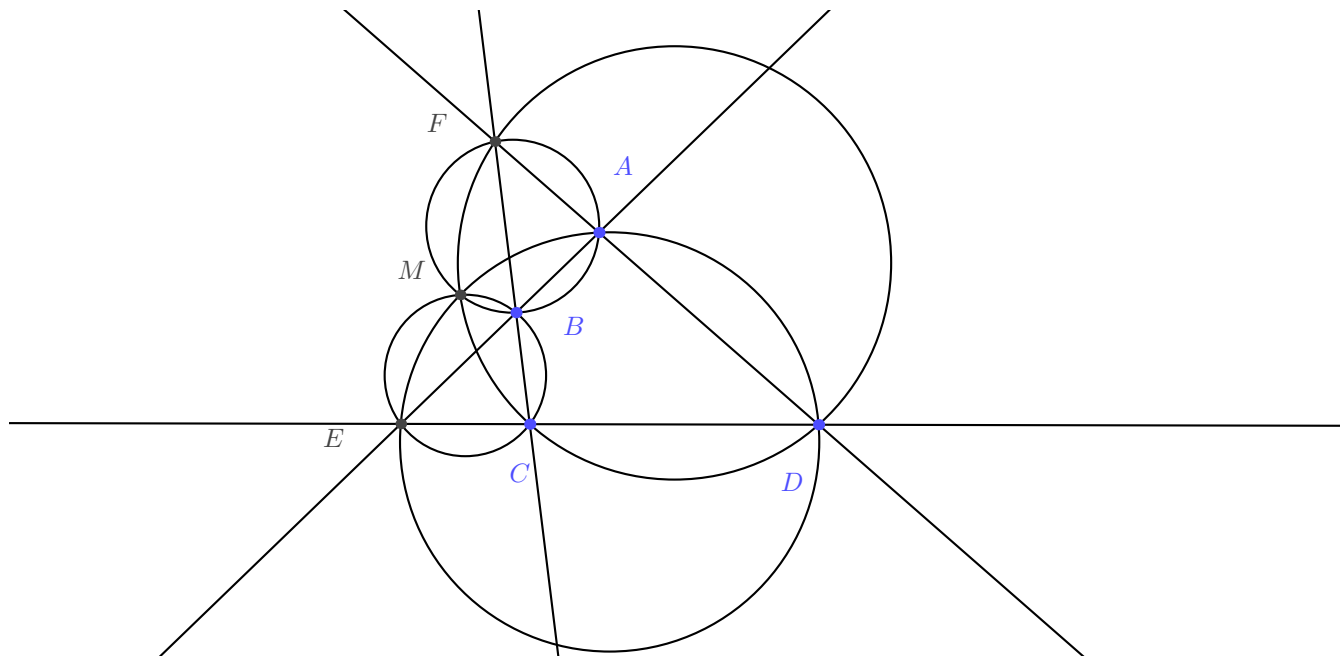


Figure 2: The magical Miquel Diagram

Try showing this theorem using angle chase or by combining Lemma 1.1, Lemma 1.3.

§2 Some Interesting Helpful Theorems

Theorem 2.1 (Feet of altitudes)

The feet of altitude from M to the 4 sides of the complete quadrilateral are collinear.

Theorem 2.2 (Gauss-Bodenmiller)

The circles with diameter AC, BD, EF are coaxial and the orthocenters of EBC, EAD, FAB, FCD lie on their radical axis. Their radical axis is known as the Aubert Line and this line is perpendicular to the line in Theorem 2.1.

Theorem 2.3

The centers of $(EBC), (EAD), (FAB), (FCD)$ and M are concyclic.

§3 The cyclic Miquel

Now, assume that the quadrilateral $ABCD$ is cyclic with center O circumcircle ω and diagonals AC, BD intersecting at K . Now, we have the following results.

Theorem 3.1 (Cyclic Miquel)

Some magical results additional to the above results.

1. M is the inverse of K with respect to ω .
2. M lies on $\overline{EF}$ and $OM \perp EF$.
3. B, M, O, D are concyclic.
4. C, M, O, A are concyclic.
5. MO bisects $\angle BMD$ and $\angle AMC$.
6. Find more yourself :d

I'll also mention the following theorem as it seems important not to mention.

Theorem 3.2 (Brokard's Theorem)

The triangle EFK is self-polar (i.e. K is pole of EF , E is pole of FK and F is pole of EK) and has orthocenter O .

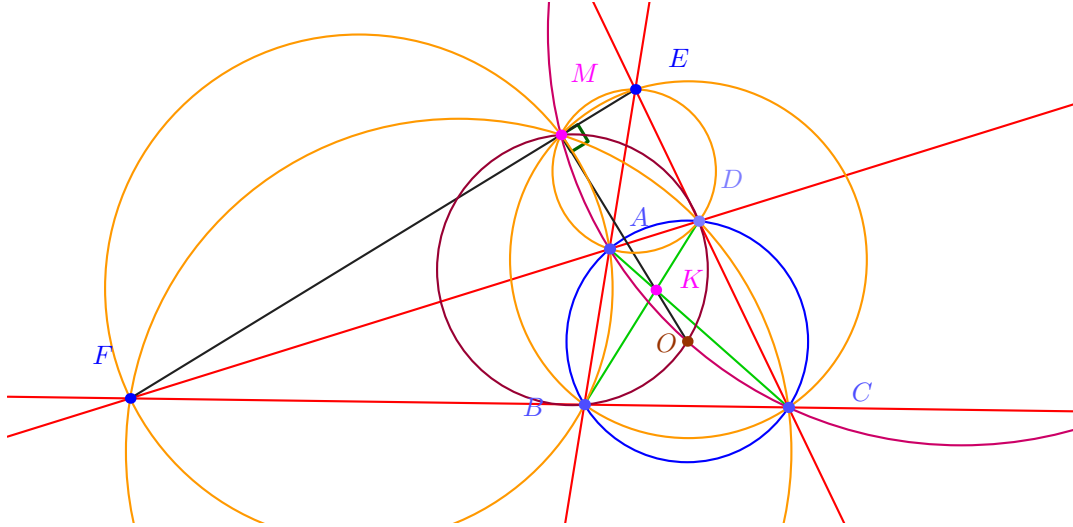


Figure 3: The magical cyclic Miquel figure

Now, we will try some actual contest problems to get a taste of the uses of Spiral Similarity.

§4 Example Problems

Example 4.1 (Brazil 2011/5)

Let ABC be an acute triangle and H is orthocenter. Let D be the intersection of BH and AC and E be the intersection of CH and AB . The circumcircle of ADE cuts the circumcircle of ABC at $F \neq A$. Prove that the angle bisectors of $\angle BFC$ and $\angle BHC$ concur at a point on BC .

Proof. The problem is equivalent to showing that $\frac{BF}{CF} = \frac{BH}{CH}$. But, we know that $BEH \sim CED$, thus $\frac{BH}{CH} = \frac{BE}{CE}$. Now, observe that F is Miquel point of $BEDC$, thus it is the center of spiral similarity that takes $BE \mapsto CD$, thus $\triangle BEF \sim \triangle FCD$, thus, $\frac{FB}{FC} = \frac{EB}{DC} = \frac{BH}{HC}$ and we are done.

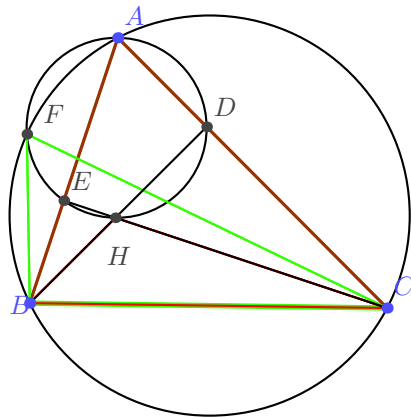


Figure 4: Brazil 2011/5

□

Example 4.2 (USAMO 2006/6)

Let $ABCD$ be a quadrilateral, and let E and F be points on sides AD and BC , respectively, such that $\frac{AE}{ED} = \frac{BF}{FC}$. Ray FE meets rays BA and CD at S and T , respectively. Prove that the circumcircles of triangles SAE , SBF , TCF , and TDE pass through a common point.

Proof. Let M be the Miquel point of quadrilateral $ABCD$, then we must have that a spiral similarity centered at M maps $AD \mapsto BC$ but we have $\frac{AE}{ED} = \frac{BF}{FC}$, thus this spiral similarity also takes E to F . Thus, M is also the spiral center of $FEDC$ and $FEAB$. Thus, M must lie on all the mentioned circles.

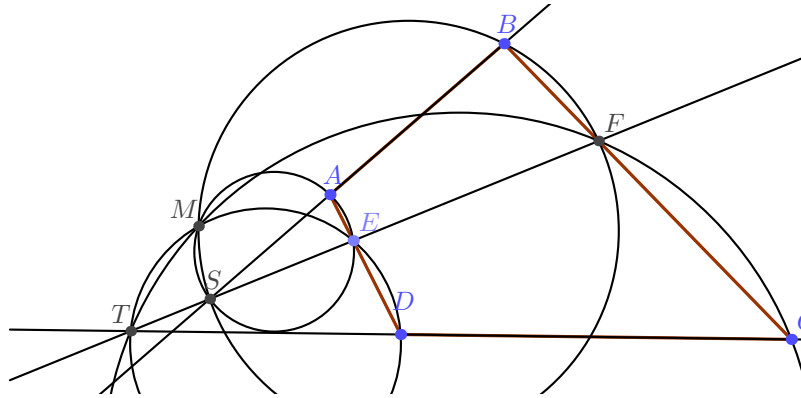


Figure 5: USAMO 2006/6

□

Example 4.3 (ISL 1995/G8)

Suppose that $ABCD$ is a cyclic quadrilateral. Let $E = AC \cap BD$ and $F = AB \cap CD$. Denote by H_1 and H_2 the orthocenters of triangles EAD and EBC , respectively. Prove that the points F , H_1 , H_2 are collinear.

Proof. Consider the polygon $ACBD$. Then, by Gauss-Bodenmiller Theorem, we have that the line H_1, H_2 is the radical axis of the circles with diameters AB, DC . Thus, we need to show that F has equal powers with both these circles. But, $FA \cdot FB = FD \cdot DC$ as $ABCD$ is cyclic so we are done. See Fig. 6 for reference.

□

Clearly, Spiral Similarity is quite powerful and the Miquel point configuration is one of the most beautiful ones.

Now, in the next section, you can try some of these problems for some real contest practice.

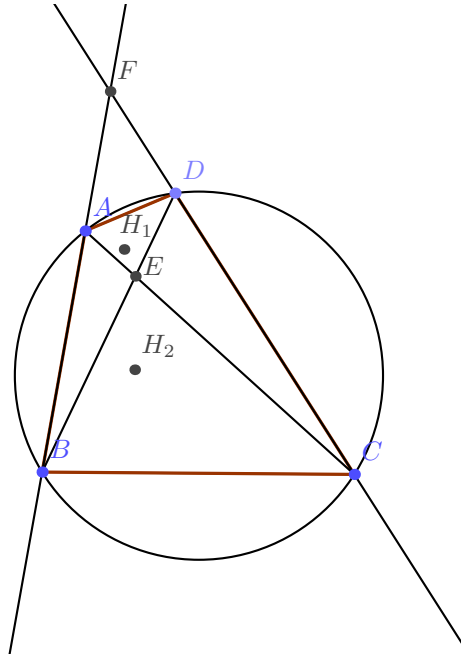


Figure 6: ISL 1995/G8

§5 Contest Practice

Problem 5.1 (USEMO 2019?/1). Let $ABCD$ be a cyclic quadrilateral. A circle centered at O passes through B and D and meets lines BA and BC again at points E and F (distinct from A, B, C). Let H denote the orthocenter of triangle DEF . Prove that if lines AC, DO, EF are concurrent, then triangles ABC and EHF are similar.

Problem 5.2 (IMO 1985/5). A circle with center O passes through the vertices A and C of the triangle ABC and intersects the segments AB and BC again at distinct points K and N respectively. Let M be the point of intersection of the circumcircles of triangles ABC and KBN (apart from B). Prove that $\angle OMB = 90^\circ$.

Problem 5.3 (APMO 2017/2). Let ABC be a triangle with $AB < AC$. Let D be the intersection point of the internal bisector of angle BAC and the circumcircle of ABC . Let Z be the intersection point of the perpendicular bisector of AC with the external bisector of angle $\angle BAC$. Prove that the midpoint of the segment AB lies on the circumcircle of triangle ADZ .

Problem 5.4 (USA EGMO/TST 2020/2). Let ABC be a triangle and let P be a point not lying on any of the three lines AB , BC , or CA . Distinct points D , E , and F lie on lines BC , AC , and AB , respectively, such that $\overline{DE} \parallel \overline{CP}$ and $\overline{DF} \parallel \overline{BP}$. Show that there exists a point Q on the circumcircle of $\triangle AEF$ such that $\triangle BAQ$ is similar to $\triangle PAC$.

Problem 5.5 (ISL 2005 G5). Let $\triangle ABC$ be an acute-angled triangle with $AB \neq AC$. Let H be the orthocenter of triangle ABC , and let M be the midpoint of the side BC . Let D be a point on the side AB and E a point on the side AC such that $AE = AD$ and the points D, H, E are on the same line. Prove that the line HM is perpendicular to the common chord of the circumscribed circles of triangle $\triangle ABC$ and triangle $\triangle ADE$.

Problem 5.6 (USATSTST 2012/2). Let $ABCD$ be a quadrilateral with $AC = BD$. Diagonals AC and BD meet at P . Let ω_1 and O_1 denote the circumcircle and the circumcenter of triangle ABP . Let ω_2 and O_2 denote the circumcircle and circumcenter of triangle CDP . Segment BC meets ω_1 and ω_2 again at S and T (other than B and C), respectively. Let M and N be the midpoints of minor arcs $\widehat{SP}$ (not including B) and $\widehat{TP}$ (not including C). Prove that $MN \parallel O_1O_2$.

Problem 5.7 (ISL 2015 G3). Let ABC be a triangle with $\angle C = 90^\circ$, and let H be the foot of the altitude from C . A point D is chosen inside the triangle CBH so that CH bisects AD . Let P be the intersection point of the lines BD and CH . Let ω be the semicircle with diameter BD that meets the segment CB at an interior point. A line through P is tangent to ω at Q . Prove that the lines CQ and AD meet on ω .

Problem 5.8 (USATST 2013). Let ABC be a scalene triangle with $\angle BCA = 90^\circ$, and let D be the foot of the altitude from C . Let X be a point in the interior of the segment CD . Let K be the point on the segment AX such that $BK = BC$. Similarly, let L be the point on the segment BX such that $AL = AC$. The circumcircle of triangle DKL intersects segment AB at a second point T (other than D). Prove that $\angle ACT = \angle BCT$.

Problem 5.9 (USAMO 2018/5). In convex cyclic quadrilateral $ABCD$, we know that lines AC and BD intersect at E , lines AB and CD intersect at F , and lines BC and DA intersect at G . Suppose that the circumcircle of $\triangle ABE$ intersects line CB at B and P , and the circumcircle of $\triangle ADE$ intersects line CD at D and Q , where C, B, P, G and C, Q, D, F are collinear in that order. Prove that if lines FP and GQ intersect at M , then $\angle MAC = 90^\circ$.

Problem 5.10 (USATST 2017/2). Let ABC be an acute scalene triangle with circumcenter O , and let T be on line BC such that $\angle TAO = 90^\circ$. The circle with diameter $\overline{AT}$ intersects the circumcircle of $\triangle BOC$ at two points A_1 and A_2 , where $OA_1 < OA_2$. Points B_1, B_2, C_1, C_2 are defined analogously.

- Prove that $\overline{AA_1}, \overline{BB_1}, \overline{CC_1}$ are concurrent.
- Prove that $\overline{AA_2}, \overline{BB_2}, \overline{CC_2}$ are concurrent on the Euler line of triangle ABC .

Problem 5.11 (USATST 2007/5). Triangle ABC is inscribed in circle ω . The tangent lines to ω at B and C meet at T . Point S lies on ray BC such that $AS \perp AT$. Points B_1 and C_1 lie on ray ST (with C_1 in between B_1 and S) such that $B_1T = BT = C_1T$. Prove that triangles ABC and AB_1C_1 are similar to each other.

Problem 5.12 (ELMO 2010/6). Let ABC be a triangle with circumcircle ω , incenter I , and A -excenter I_A . Let the incircle and the A -excircle hit BC at D and E , respectively, and let M be the midpoint of arc BC without A . Consider the circle tangent to BC at D and arc BAC at T . If TI intersects ω again at S , prove that SI_A and ME meet on ω .

Problem 5.13 (ISL 2006 G9). Points A_1, B_1, C_1 are chosen on the sides BC, CA, AB of a triangle ABC respectively. The circumcircles of triangles $AB_1C_1, BC_1A_1, CA_1B_1$ intersect the circumcircle of triangle ABC again at points A_2, B_2, C_2 respectively ($A_2 \neq A, B_2 \neq B, C_2 \neq C$). Points A_3, B_3, C_3 are symmetric to A_1, B_1, C_1 with respect to the midpoints of the sides BC, CA, AB respectively. Prove that the triangles $A_2B_2C_2$ and $A_3B_3C_3$ are similar.

§6 Post-Lecture

If you haven't already, I highly recommend going through Yufei Zhao's handouts on geometry especially- http://yufeizhao.com/olympiad/cyclic_quad.pdf. Other handouts can also be found at <http://yufeizhao.com/olympiad/>. Evan's book EGMO (Euclidian Geometry in Mathematical Olympiads) is one of the best geometry intro resources out there and his website has an immense collection of great resources as well. Find them at <https://web.evanchen.cc/>. His book and Yufei's notes were taken as references in writing this handout.